

## PEACHTREE CITY, GA, USA

WMO: 722197

Lat: 33.355N

Lon: 84.567W

Elev: 798

StdP: 14.28

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 53819

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |            |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|------------|------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |            |             | 99%        |            |             | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP         | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>20.0 | (c)<br>24.1 | (d)<br>5.8                    | (e)<br>7.6 | (f)<br>30.2 | (g)<br>9.6 | (h)<br>9.3 | (i)<br>33.3 | (j)<br>20.7           | (k)<br>38.1 | (l)<br>18.7 | (m)<br>40.8 | (n)<br>4.5            | (o)<br>320 | (p)<br>0.364 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.9                   | 93.7            | 74.1 | 91.8 | 73.9 | 90.1 | 73.7 | 77.8                | 87.3 | 76.9 | 86.4 | 76.0 | 85.1 | 6.1                  | 320  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 75.2                            | 136.3 | 81.3 | 74.2 | 131.5 | 80.3 | 73.3 | 127.3 | 79.4 | 41.7          | 87.4 | 40.7 | 86.4 | 39.9 | 85.1 | 82.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |       | n=20 years |       | n=50 years |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 17.3              | 14.9 | 12.3 | 14.0                       | 96.5 | 4.2                | 3.4 | 11.0                                               | 98.9 | 8.5        | 100.8 | 6.1        | 102.7 | 3.1        | 105.2 |
|                   |      |      | 12.4                       | 79.8 | 4.3                | 1.6 | 9.3                                                | 80.9 | 6.8        | 81.9  | 4.4        | 82.8  | 1.2        | 83.9  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |       |            |       |            |       |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 61.7   | 43.4  | 47.0  | 53.6  | 60.9  | 69.1  | 76.0  | 78.9  | 78.3  | 72.7  | 62.2  | 51.7  | 46.0  |
|                                                                        | DBStd       | 14.46  | 9.64  | 8.84  | 8.50  | 6.85  | 5.97  | 4.30  | 3.03  | 3.53  | 5.46  | 7.71  | 7.72  | 8.77  |
|                                                                        | HDD50       | 738    | 254   | 150   | 63    | 6     | 0     | 0     | 0     | 0     | 0     | 7     | 70    | 188   |
|                                                                        | HDD65       | 2872   | 671   | 505   | 362   | 156   | 33    | 1     | 0     | 0     | 9     | 143   | 403   | 589   |
|                                                                        | CDD50       | 5019   | 48    | 66    | 174   | 334   | 592   | 779   | 896   | 877   | 682   | 386   | 120   | 65    |
|                                                                        | CDD65       | 1679   | 0     | 2     | 7     | 34    | 160   | 331   | 431   | 412   | 241   | 57    | 3     | 1     |
|                                                                        | CDH74       | 15400  | 0     | 11    | 118   | 470   | 1609  | 3057  | 3984  | 3637  | 1979  | 505   | 28    | 2     |
|                                                                        | CDH80       | 6243   | 0     | 0     | 12    | 80    | 520   | 1336  | 1803  | 1629  | 760   | 102   | 1     | 0     |
| (14) Wind                                                              | WSAvg       | 4.0    | 5.3   | 5.2   | 5.4   | 4.8   | 3.8   | 3.2   | 3.1   | 2.7   | 2.9   | 3.4   | 4.1   | 4.6   |
| (15) Precipitation                                                     | PrecAvg     | 48.0   | 4.2   | 4.4   | 5.2   | 3.7   | 3.7   | 4.2   | 4.5   | 3.5   | 3.6   | 2.8   | 3.7   | 4.4   |
|                                                                        | PrecMax     | 68.6   | 6.1   | 8.1   | 10.3  | 7.5   | 8.9   | 9.6   | 14.4  | 6.3   | 10.9  | 8.6   | 10.0  | 12.8  |
|                                                                        | PrecMin     | 33.1   | 2.9   | 1.8   | 1.0   | 1.5   | 1.3   | 2.0   | 2.2   | 1.1   | 0.3   | 0.1   | 0.9   | 1.2   |
|                                                                        | PrecStd     | 10.1   | 1.0   | 1.6   | 2.4   | 1.5   | 1.8   | 2.2   | 2.9   | 1.3   | 2.5   | 2.0   | 2.1   | 2.8   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 71.9  | 75.6  | 81.6  | 85.7  | 91.3  | 95.2  | 96.0  | 97.0  | 94.6  | 87.0  | 78.6  | 73.1  |
|                                                                        |             | MCWB   | 60.5  | 62.4  | 62.8  | 65.7  | 71.1  | 73.7  | 74.6  | 74.3  | 72.9  | 72.0  | 64.8  | 63.3  |
|                                                                        | 2%          | DB     | 67.4  | 70.4  | 77.3  | 82.2  | 88.2  | 92.7  | 93.1  | 93.3  | 90.7  | 82.3  | 73.3  | 69.9  |
|                                                                        |             | MCWB   | 59.7  | 59.4  | 61.5  | 64.9  | 70.2  | 73.3  | 74.7  | 74.8  | 71.9  | 67.9  | 61.0  | 62.5  |
|                                                                        | 5%          | DB     | 63.9  | 66.8  | 73.4  | 79.6  | 85.3  | 90.4  | 91.0  | 90.9  | 87.8  | 79.5  | 70.6  | 65.6  |
|                                                                        |             | MCWB   | 57.7  | 57.4  | 59.1  | 63.3  | 68.7  | 72.9  | 75.0  | 74.4  | 72.1  | 65.5  | 60.4  | 60.2  |
|                                                                        | 10%         | DB     | 60.6  | 63.5  | 70.0  | 76.3  | 82.6  | 87.9  | 89.3  | 88.5  | 84.3  | 76.1  | 66.9  | 62.1  |
|                                                                        |             | MCWB   | 55.0  | 55.5  | 57.7  | 61.7  | 68.2  | 72.5  | 74.9  | 74.2  | 71.1  | 64.8  | 58.3  | 56.6  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 64.5  | 65.4  | 66.4  | 70.5  | 74.8  | 78.0  | 79.2  | 78.9  | 76.5  | 74.9  | 69.5  | 67.5  |
|                                                                        |             | MCDB   | 67.0  | 70.4  | 74.2  | 78.6  | 83.2  | 87.5  | 87.9  | 88.1  | 85.5  | 81.9  | 72.2  | 69.6  |
|                                                                        | 2%          | WB     | 62.1  | 63.2  | 64.5  | 68.4  | 73.1  | 76.5  | 78.0  | 77.7  | 75.2  | 72.1  | 66.1  | 64.2  |
|                                                                        |             | MCDB   | 65.5  | 67.7  | 72.0  | 76.4  | 82.3  | 86.6  | 87.5  | 87.1  | 84.0  | 78.2  | 70.0  | 67.1  |
|                                                                        | 5%          | WB     | 59.5  | 60.6  | 62.7  | 66.8  | 71.7  | 75.3  | 77.1  | 76.7  | 74.2  | 70.0  | 63.4  | 61.5  |
|                                                                        |             | MCDB   | 63.3  | 65.2  | 70.0  | 74.8  | 81.1  | 85.4  | 86.7  | 85.8  | 82.5  | 75.5  | 66.8  | 64.7  |
|                                                                        | 10%         | WB     | 55.7  | 57.6  | 60.3  | 65.0  | 70.5  | 74.3  | 76.1  | 75.7  | 73.1  | 67.9  | 60.4  | 57.6  |
|                                                                        |             | MCDB   | 58.8  | 62.1  | 67.2  | 72.4  | 79.3  | 83.6  | 85.3  | 84.3  | 80.7  | 73.4  | 65.6  | 60.9  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 21.7  | 22.9  | 24.4  | 25.2  | 23.8  | 21.1  | 19.9  | 19.8  | 21.3  | 24.2  | 24.7  | 21.5  |
|                                                                        |             | MCDBR  | 25.6  | 26.8  | 29.4  | 29.4  | 26.7  | 25.1  | 22.9  | 23.6  | 25.7  | 27.2  | 28.4  | 24.2  |
|                                                                        | 5% WB       | MCWBR  | 17.8  | 17.0  | 14.9  | 13.6  | 11.3  | 8.9   | 8.1   | 8.2   | 10.0  | 13.6  | 17.2  | 17.3  |
|                                                                        |             | MCWBR  | 20.2  | 21.3  | 22.9  | 22.1  | 22.0  | 20.6  | 19.6  | 19.5  | 20.2  | 19.5  | 20.5  | 20.3  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.316 | 0.320 | 0.352 | 0.387 | 0.442 | 0.476 | 0.518 | 0.520 | 0.421 | 0.368 | 0.340 | 0.317 |
|                                                                        | taud        |        | 2.524 | 2.508 | 2.441 | 2.323 | 2.216 | 2.172 | 2.065 | 2.047 | 2.313 | 2.450 | 2.471 | 2.538 |
|                                                                        | Ebn at Noon |        | 286   | 297   | 293   | 286   | 270   | 259   | 248   | 245   | 267   | 275   | 274   | 279   |
|                                                                        | Edh at Noon |        | 27    | 30    | 34    | 40    | 46    | 48    | 53    | 53    | 39    | 31    | 28    | 25    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 867   | 1048  | 1394  | 1756  | 1938  | 1949  | 1885  | 1724  | 1502  | 1278  | 963   | 737   |
|                                                                        | RadStd      |        | 49    | 128   | 119   | 117   | 162   | 162   | 130   | 101   | 148   | 162   | 96    | 77    |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | +1.82 | N/A     | N/A    | N/A     | N/A   | N/A   | -203        | -418  | +635  | +378  |
| (47) Regional (39 neighbors) |  | +0.81 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +261  | +189  |

Nomenclature: See separate page